

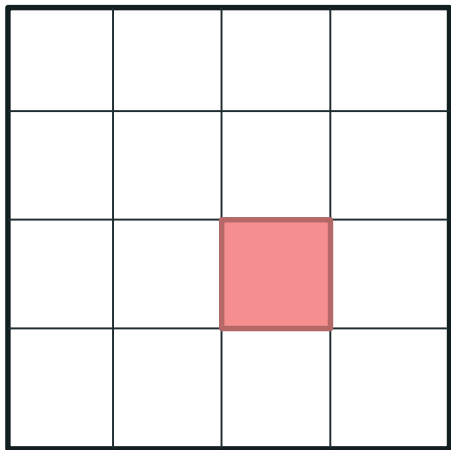
Voxel Field Fusion for 3D Object Detection

Yanwei Li, Xiaojuan Qi, Yukang Chen, Liwei Wang, Zeming Li, Jian Sun, Jiaya Jia
CVPR(2022)

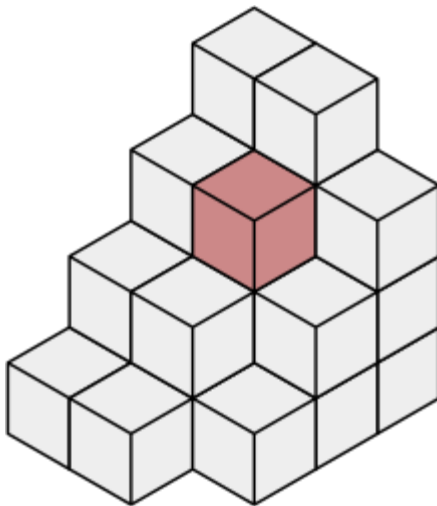
이미지처리팀(Detection)

김병현, 김선옥, 김현진, 안종식, 이찬혁, 현청천

Voxel field



2D Pixel

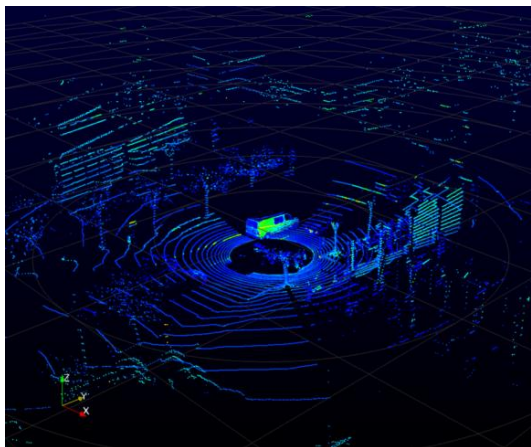
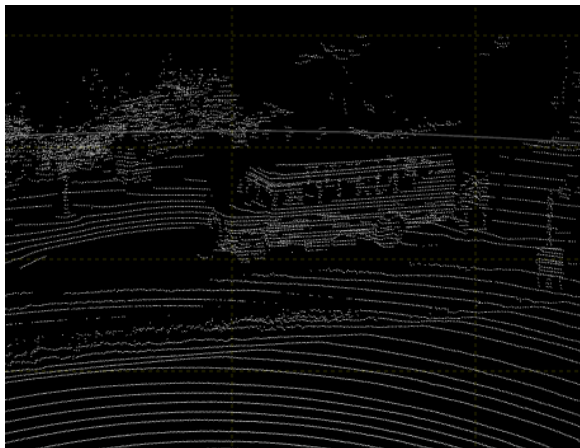


3D Voxel

Voxel = Volume + Pixel
→ 부피를 가진 픽셀

Point cloud data

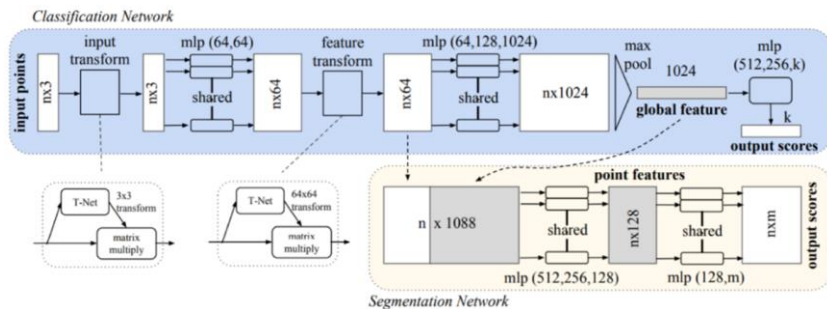
- 3차원 공간 정보를 시각적으로 표현해 주는 데이터
- 2D 데이터와 달리 정형화 되어있지 않음
- 공간상 수많은 점들을 순서없이 기록하는 방식 → Sparse, point간 상호작용 등 기하학적 특성 파악이 어려움



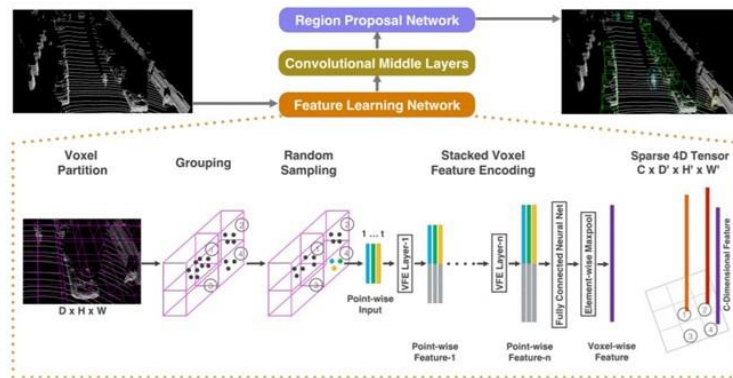
X	Y	Z	intensity
0.003	3.926	-2.328	8
0.004	4.152	-2.333	5
0.005	4.406	-2.343	3
0.018	17.237	-2.016	2
0.006	4.679	-2.350	4
0.022	17.793	-1.660	2
0.007	4.942	-2.339	4
0.008	5.271	-2.347	4
0.010	5.640	-2.356	5
0.012	6.064	-2.368	3
0.012	5.169	-1.747	1

3D AI model using Point cloud

- Point cloud data가 가진 성질의 한계를 극복하는 방향으로 관련 연구가 수행됨
→ data로부터 유의미한 정보 추출
- PointNet, VoxelNet 등



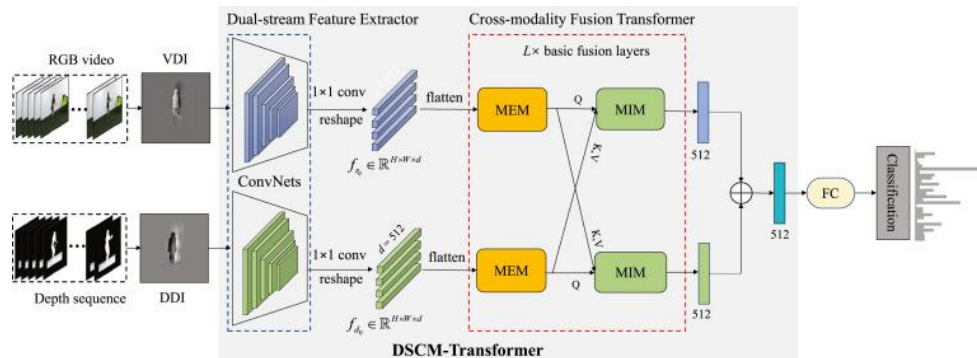
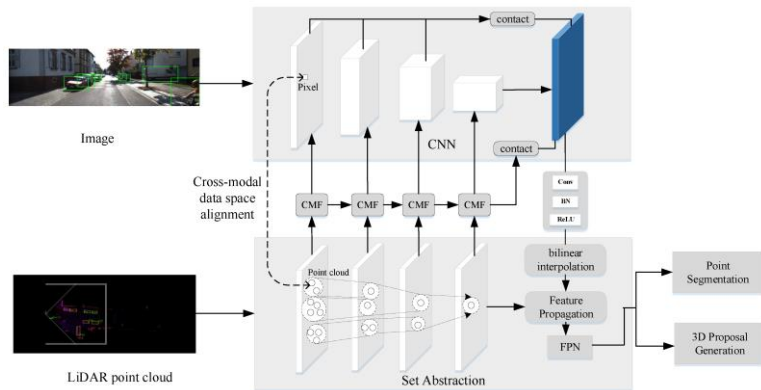
PointNet



VoxelNet

Cross-modality fusion

- Image-Point cloud, RGB-Depth, Image-Word 등 다른 양식(modality)을 갖는 데이터를 융합하는 것
- 서로 다른 modality에서 오는 정보의 양이 unbalance하고 feature space또한 unbalance함
audio – waveform, image-2D array, text-word embedding vector

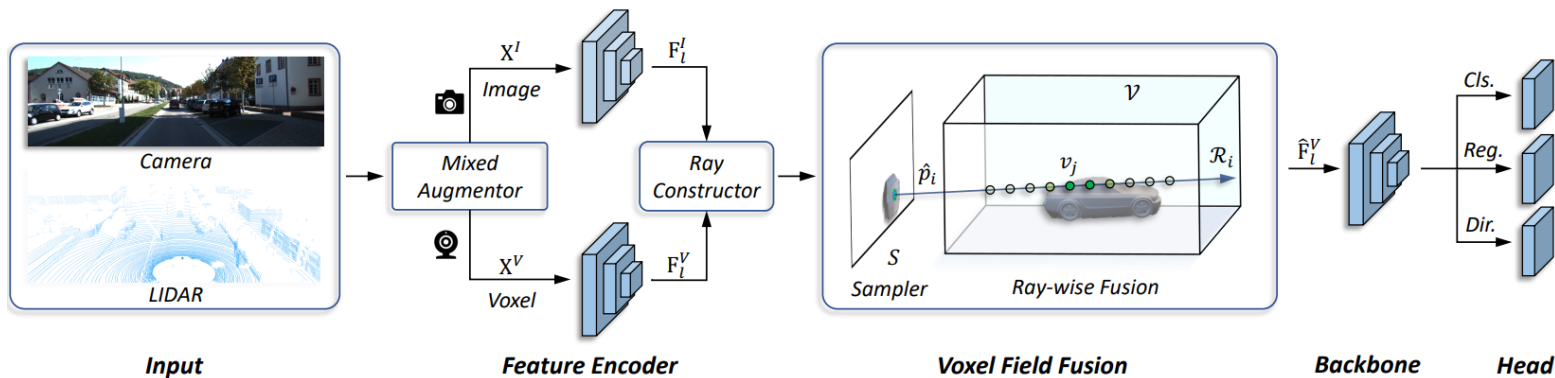




Question

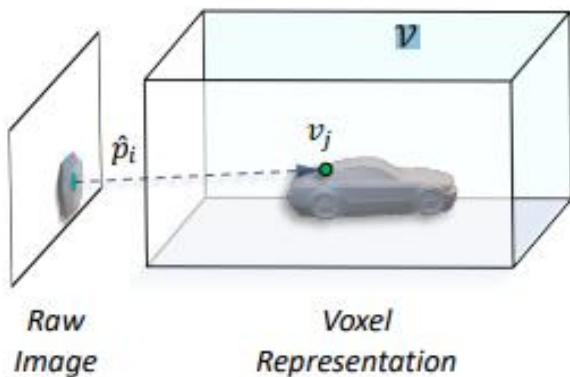
Purpose of the paper

- Cross-modality 3D 객체인식을 위한 효율적인 framework 제시
- LiDAR와 RGB이미지에서 얻은 값들을 어떻게 blending하는 방법 제안
- Voxel field에 augmented image feature를 ray 형식으로 나타내서 cross-modality consistency를 유지

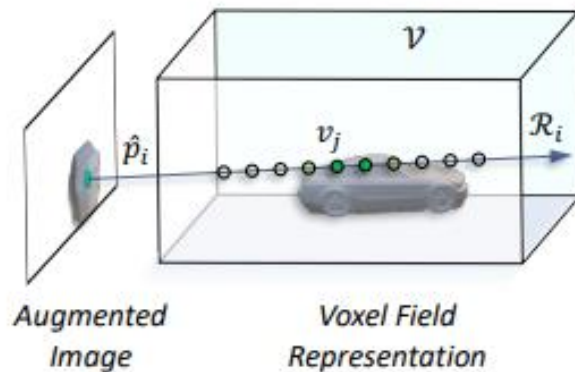


Voxel Field Fusion (VFF)

- A new cross-modality framework
- Image feature를 point-to-ray 방식으로 voxel field에 투영
- Augmented image에서 효과적으로 high-responed feature를 샘플링함



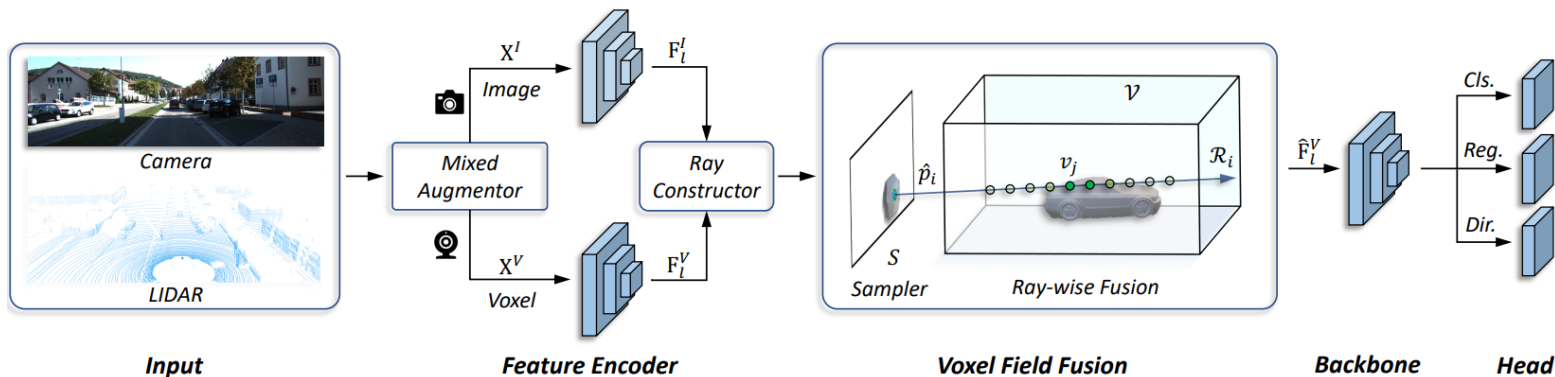
(a) Point-to-point manner



(b) Point-to-ray manner

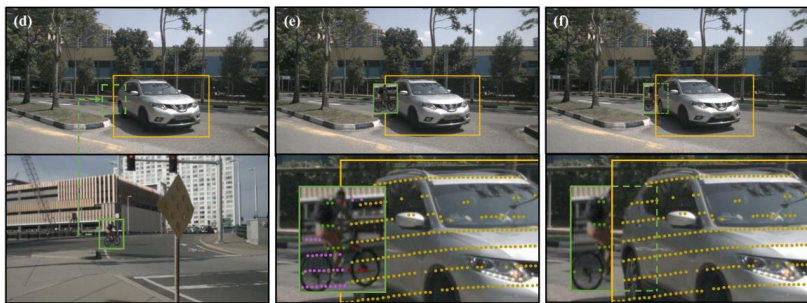
Voxel Field Fusion (VFF)

- Mixed augmentor: data modality가 달라도 data augmentation이 가능하도록 align시킴
- Learnable sampler: interaction에서 효과적으로 key feature를 선택할 수 있도록 함
- Ray-wise fusion: ray를 따라 있는 feature를 fuse하고 combine함



Mixed augmentor

- Image와 point cloud데이터가 주어진다면 데이터들을 correspondence하게 만드는 단계
- Sample-added & Sample-static augmentation 방식 사용
- Sample-added
전체 DB의 각 scene안의 sample를 증가:
Point-cloud data - GroundTruth-Sampling
Image data - copy-paste
- Sample-static
새로운 sample를 추가하지않고 transformation
2D conv에 영향을 주도록 image-level operation사용



Sample-added

<i>type</i>	Point operation	Image operation
Sample-added	GT-sampling	Copy-paste
Sample-static	Flip Rescale Rotate	Image-flip Image-rescale Reproject

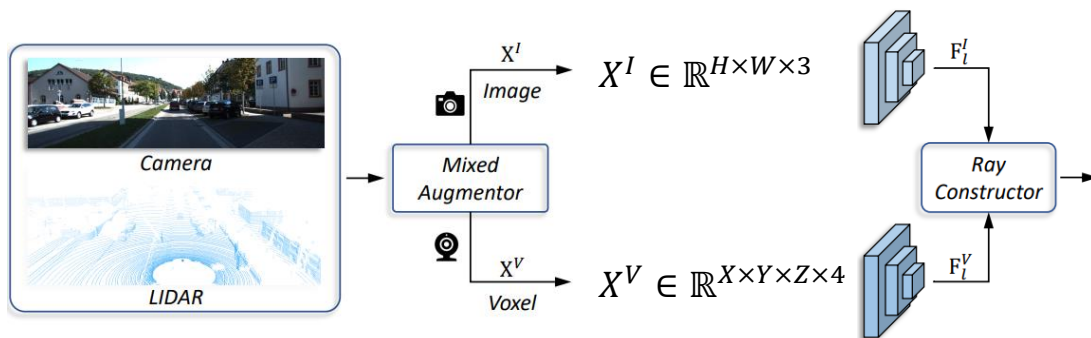
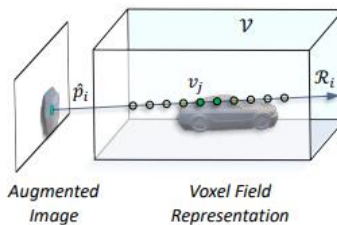
Voxel Field Construction

- Voxel Field
voxel bin v_i & image pixel $p_i \rightarrow$ given projection matrix $T_{Voxel \rightarrow Image}$
- Single voxel v 는 함수 F 로 표현 가능함

$$v \text{ with } (x, y, z), F_{l,v}^V = F(x, y, z)$$

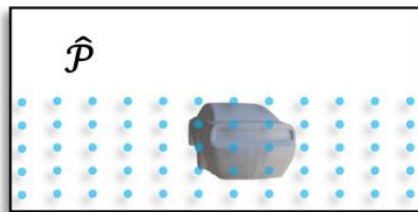
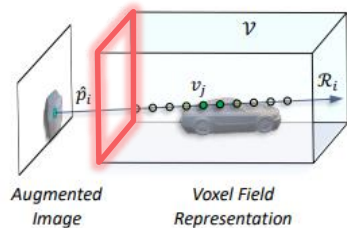
- Image pixel point p_i 로부터 나오는 ray $\mathcal{R}_i$

$$p_i = v_j T_{Voxel \rightarrow Image}^T, \forall v_j \in \mathcal{R}_i$$

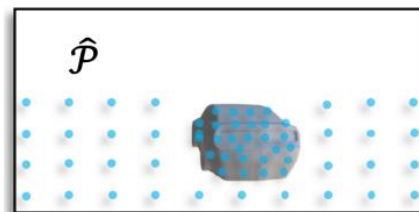


Learnable Sampler

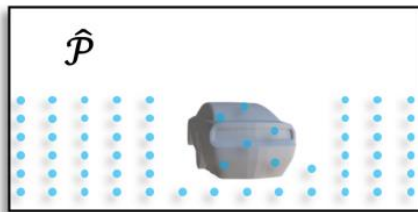
- Computational burden을 줄이기 위해 ray interaction의 **image plane**에서 n개의 points만 선택
- $\mathcal{S}(\mathcal{P}) = \mathcal{U}(\{p_i : \mathbb{1}(\delta(f(F_{l,i}^I))) = 1\})$ 중요도를 어떻게 산출하는지?
Function f



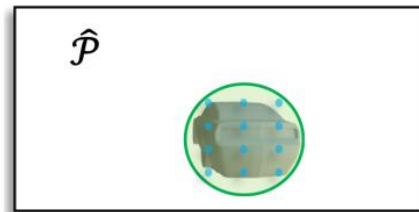
(a) Sample by uniformity



(b) Sample by density



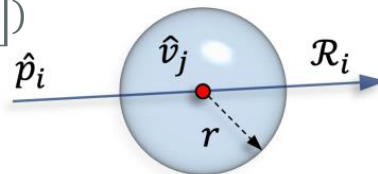
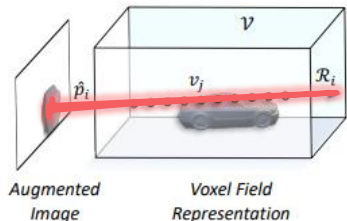
(c) Sample by sparsity



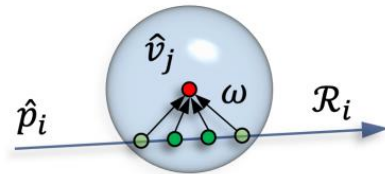
(d) Sample by importance

Ray-voxel Interaction

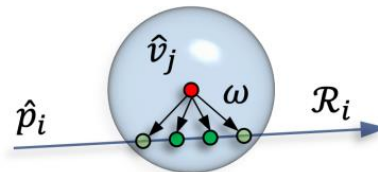
- Designed local fusion extends the perception region from a single voxel to near area
- Operation region을 whole ray로 확장함 → ray-wise fusion
- 하나의 ray에서 몇 개의 점을 선택할 것인지에 대한 방법 제시
- $\omega_j = \delta(< F_{l,i}^I, F'_{l,v_j} >)$
- $\hat{\mathcal{F}}(x_j, y_j, z_j) = \mathcal{F}(x_j, y_j, z_j) + \omega_j f([F_{l,i}^I, F'_{l,v_j}])$



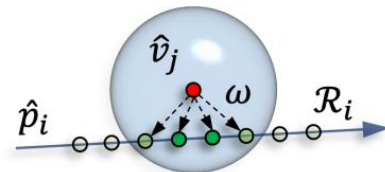
(a) Single fusion with each point



(b) Local fusion with aggregation



(c) Local fusion with propagation



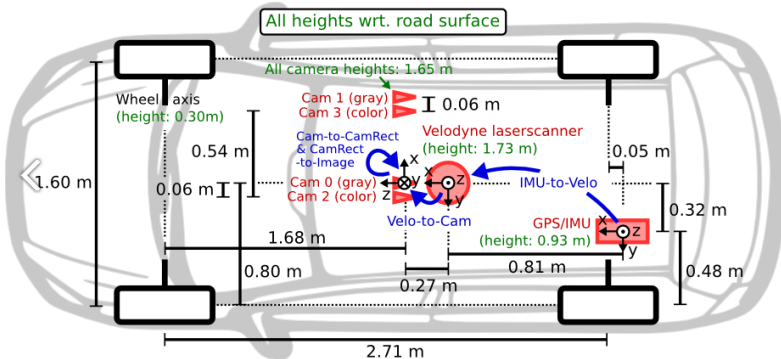
(d) Ray-wise fusion along the ray



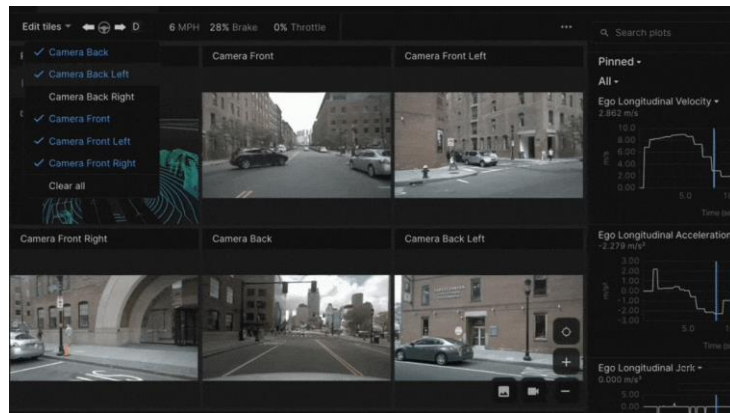
Question

Experiment setup

- KITTI dataset with PV-RCNN(backbone)
- 비교군: nuScenes, KITTI dataset
- KITTI dataset: 자율주행을 위한 대표적인 데이터셋 (synced LiDAR point and camera images)
- nuScenes dataset: large-scale benchmark (synced data with 10 object categories, 32-beam LiDAR and 6 cameras)



KITTI dataset



nuScenes dataset

Experiment setup

- PV-RCNN: Point voxel RCNN

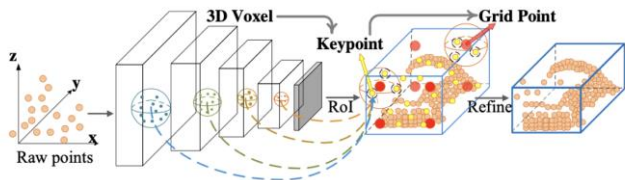
2-stage detection구조로, box proposal → confidence & location refinement
grid 방식과 point 방식을 integration해서 기존 backbone 대비 큰 성능향상을 이룸

- Voxel R-CNN

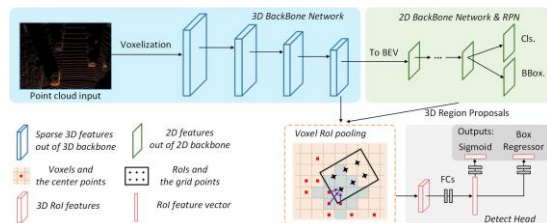
2-stage detection구조, voxel feature를 최대한 활용하도록 설계됨, coarse한 데이터를 사용해도 충분히 좋은 성능의 3D detector를 만들 수 있음 보임
3D backbone Network, 2D bird-eye-view(BEV), RPN, Detection head로 구성됨

- CenterPoint

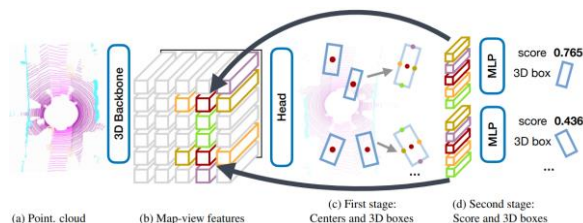
LiDAR point clouds로 feature extraction → object center를 2D CNN 구조로 detection하는 구조



PV-RCNN



Voxel R-CNN



CenterPoint

Main Conclusion – KITTI dataset

- PV-RCNN과 Voxel R-CNN에 적용했을 때, 성능향상을 보임 (validation set)
- Test set에 수행한 다른 method과 비교해도 81 AP이상 높은 성능을 보임

Table 8. Comparisons on different methods with a single model on the KITTI *val* set. * denotes our result from official source code.

method	AP _{3D} @Car-R40 (IoU=0.7)			AP _{3D} @Car-R11 (IoU=0.7)			AP _{BEV} @Car-R40 (IoU=0.7)		
	Easy	Moderate	Hard	Easy	Moderate	Hard	Easy	Moderate	Hard
<i>LiDAR-based</i>									
SECOND [38]	–	–	–	87.43	76.48	69.10	–	–	–
PointRCNN [28]	–	–	–	88.88	78.63	77.38	–	–	–
STD [42]	–	–	–	89.70	79.80	79.30	–	–	–
PV-RCNN [27]	92.57	84.83	82.69	89.35	83.69	78.70	95.76	91.11	88.93
Voxel R-CNN [8]	92.38	85.29	82.86	89.41	84.52	78.93	95.52	91.25	88.99
<i>LiDAR+RGB</i>									
UberATG-MMF [16]	–	–	–	88.40	77.43	70.22	–	–	–
3D-CVF [46]	89.67	79.88	78.47	–	–	–	–	–	–
EPNet [13]	92.28	82.59	80.14	–	–	–	95.51	88.76	88.36
PV-RCNN*	91.53	84.36	82.29	88.95	83.51	78.72	92.82	90.43	88.41
+ VFF	92.31	85.51	82.92	89.45	84.21	79.13	95.43	91.40	90.66
Voxel R-CNN*	92.27	84.88	82.50	89.46	83.61	78.80	95.51	91.13	88.85
+ VFF	92.47	85.65	83.38	89.51	84.76	79.21	95.65	91.75	91.39

Table 13. Comparisons on different methods on KITTI *test* set.

method	AP _{3D} @Car-R40 (IoU=0.7)		
	Easy	Moderate	Hard
<i>LiDAR-based</i>			
PointPillars [15]	82.58	74.31	68.99
PointRCNN [28]	86.96	75.64	70.70
Part-A ² [29]	87.81	78.49	73.51
STD [42]	87.95	79.71	75.09
SA-SSD [11]	88.75	79.79	74.16
PV-RCNN [27]	90.25	81.43	76.82
Voxel R-CNN [8]	90.90	81.62	77.06
<i>LiDAR+RGB</i>			
MV3D [6]	74.97	63.63	54.00
F-PointNet [23]	82.19	69.79	60.59
AVOD [14]	83.07	71.76	65.73
UberATG-MMF [16]	88.40	77.43	70.22
EPNet [13]	89.81	79.28	74.59
3D-CVF [46]	89.20	80.05	73.11
VFF + PV-RCNN	89.58	81.97	79.17
VFF + Voxel R-CNN	89.50	82.09	79.29

Main Conclusion – nuScene dataset

- Large-scale의 nuScene test set에 대한 결과
- CenterPoint backbone에 대해 test 수행
- CenterPoint 독립적일때보다 8.1 AP이상 성능 향상

Table 12. Comparisons on different methods with single model on the nuScenes *test* set.

<i>method</i>	mAP	NDS	Car	Truck	Bus	Trailer	C.V.	Ped.	Motor.	Bicycle	T.C.	Barrier
<i>LiDAR-based</i>												
PointPillars [15]	30.5	45.3	68.4	23.0	28.2	23.4	4.1	59.7	27.4	1.1	30.8	38.9
3DSSD [41]	42.6	56.4	81.2	47.2	61.4	30.5	12.6	70.2	36.0	8.6	31.1	47.9
CBGS [52]	52.8	63.3	81.1	48.5	54.9	42.9	10.5	80.1	51.5	22.3	70.9	65.7
CenterPoint [44]	60.3	67.3	85.2	53.5	63.6	56.0	20.0	84.6	59.5	30.7	78.4	71.1
<i>LiDAR+RGB</i>												
PointPainting [33]	46.4	58.1	77.9	35.8	36.2	37.3	15.8	73.3	41.5	24.1	62.4	60.2
FusionPainting [37]	66.3	70.4	86.3	58.5	66.8	59.4	27.7	87.5	71.2	51.7	84.2	70.2
MVP [45]	66.4	70.5	86.8	58.5	67.4	57.3	26.1	89.1	70.0	49.3	85.0	74.8
PointAugmenting [34]	66.8	71.0	87.5	57.3	65.2	60.7	28.0	87.9	74.3	50.9	83.6	72.6
VFF + CenterPoint	68.4	72.4	86.8	58.1	70.2	61.0	32.1	87.1	78.5	52.9	83.8	73.9

Effect of Augmentation

- Mixed augmentor의 효과 확인 → cross-modality consistency를 유지하는가
- Aligned Augmentation
image-level transformation이 feature-variant augmentation에 대해 효과적임
- Mixed Augmentation
sample-added strategy 성능향상이 높음 (6.28% AP)
sample-static의 성능향상이 그리 크지않음 → sample-added strategy only

Table 2. Comparisons on different augmentations on the KITTI *val* set. *aug type* and *align type* denote the adopted sample-static augmentation and type of cross-modality alignment, respectively.

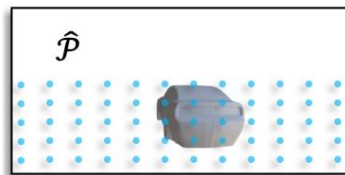
<i>aug type</i>	<i>align type</i>	AP _{3D} @Car-R40 (IoU=0.7)		
		Easy	Moderate	Hard
None	–	88.69	81.82	79.91
+ Flip	reproject	89.26	82.50	82.27
	ours	91.34	84.15	82.48
+ Rescale	reproject	91.78	84.25	82.58
	ours	91.68	84.53	82.59
+ Rotate	reproject	91.30	84.45	82.64
	ours	91.43	84.57	82.61

Table 3. Comparisons on different strategies in mixed augmentor on the KITTI *val* set. *strategy* and *fusion* denote the strategy in mixed augmentor and the usage of voxel field fusion, respectively.

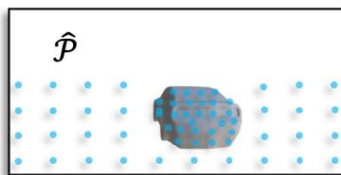
<i>strategy</i>	<i>fusion</i>	AP _{3D} @Car-R40 (IoU=0.7)		
		Easy	Moderate	Hard
Original	✗	86.40	75.47	71.32
	✓	87.08	76.25	72.03
+ Added	✗	87.77	79.19	76.71
	✓	89.56	82.53	80.00
+ Static	✗	91.53	84.36	82.29
	✓	92.31	85.51	82.92

Learnable sampler

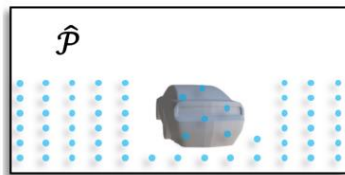
- Sampling method간 비교
- Heuristic and learnable set에 대해 비교 수행



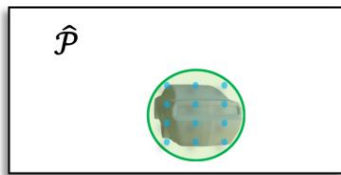
(a) Sample by uniformity



(b) Sample by density



(c) Sample by sparsity



(d) Sample by importance

Table 4. Comparisons on different sampling types on the KITTI *val* set. *sample type* and *learn* denote the adopted sampler and learnable manner for feature selection in Section 3.2, respectively.

<i>sample type</i>	<i>learn</i>	AP _{3D} @Car-R40 (IoU=0.7)		
		Easy	Moderate	Hard
Uniformity	✗	88.69	81.90	79.82
Sparsity	✗	89.09	81.68	79.85
Density	✗	88.72	82.01	81.42
Importance	✓	89.11	82.10	79.98

Ray-voxel Interaction & Ray-wise Supervision

- Ray와 voxel간 interaction이 제안한 framework의 core operation임을 확인
- Fusion method에 따른 결과(AP) 비교
- Ray-wise fusion시 gaussian distribution with radius가 어느수준이 좋은가 확인
- Gaussian ball의 radius가 커질 수록 wrong guidance를 나타냄 → radius = 1

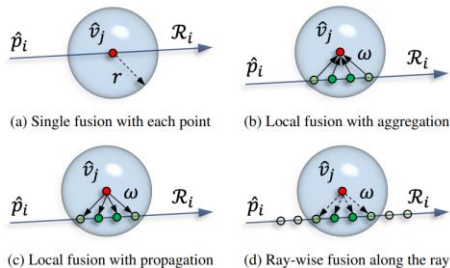


Table 5. Comparisons on different fusion strategies on the KITTI *val* set. *range* and *operation* denote the fusion range and specific operation, respectively. Single fusion is a subset of the others.

<i>range</i>	<i>operation</i>	AP _{3D} @Car-R40 (IoU=0.7)		
		Easy	Moderate	Hard
Single	–	88.69	81.82	79.91
Local	aggregate	89.00	82.12	81.52
	propagate	89.16	82.07	81.17
Ray-wise	generate	89.56	82.53	80.00

Table 6. Comparisons on different types of supervision on the KITTI *val* set. *radius* and *gaussian* denote the adopted radius r and the usage of Gaussian distribution in Section 3.3, respectively.

<i>radius</i>	<i>gaussian</i>	AP _{3D} @Car-R40 (IoU=0.7)		
		Easy	Moderate	Hard
0	✗	89.11	82.10	79.98
1	✓	89.56	82.53	80.00
2	✓	88.33	81.75	79.54



Question

Thank you